

Digital Electronics Question and Answers

Memory Devices-2

Topic 11

- 1.) Read Only Memory (ROM)
- 2.) Programmable Read Only Memory
- 3.) Erasable Programmable Read Only Memory

Digital Electronics Question & Answers

1.) Read Only Memory (ROM)

1. Which of the following has the capability to store the information permanently?

- a) RAM
- b) ROM
- c) Storage cells
- d) Both RAM and ROM

Answer: b

Explanation: ROM (Read Only Memory) has the capability to store the information permanently. RAM provides random access to memory. Storage cells are responsible for the transfer of data from and into the memory.

2. ROM has the capability to perform _____

- a) Write operation only
- b) Read operation only
- c) Both write and read operation
- d) Erase operation

Answer: b

Explanation: ROM means “Read Only Memory”. Hence, it has the capability to perform read operation only. No write or erase operation could be performed in the ROM.

3. Since, ROM has the capability to read the information only then also it has been designed, why?

- a) For controlling purpose
- b) For loading purpose
- c) For booting purpose
- d) For erasing purpose

Answer: c

Explanation: ROM means “Read Only Memory”. Hence, it has capability to perform read operation only. No write or erase operation could be performed in the ROM. It has designed to provide the computer with resident programmes and for booting purpose.

4. The ROM is a _____

- a) Sequential circuit
- b) Combinational circuit
- c) Magnetic circuit
- d) Static circuit

Answer: b

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Explanation: ROM is a combination of different ICs. So, it is a combinational circuit. It depends on present input and not past states.

5. ROM is made up of _____

- a) NAND and OR gates
- b) NOR and decoder
- c) Decoder and OR gates
- d) NAND and decoder

Answer: c

Explanation: ROM (Read Only Memory) has the capability to store the information permanently. ROM is made up of decoder and OR gates within a single IC package.

6. Why are ROMs called non-volatile memory?

- a) They lose memory when power is removed
- b) They do not lose memory when power is removed
- c) They lose memory when power is supplied
- d) They do not lose memory when power is supplied

Answer: b

Explanation: Volatile memory stores data as long as it is powered. ROMs are called non-volatile memory because of they do not lose memory when power is removed.

7. In ROM, each bit is a combination of the address variables is called _____

- a) Memory unit
- b) Storage class
- c) Data word
- d) Address

Answer: c

Explanation: In ROM, each bit is a combination of the address variables is called data word. Usually, a word consists of 16-bits or 2-bytes.

8. Which is not a removable drive?

- a) Zip
- b) Hard disk
- c) Super Disk
- d) Jaz

Answer: b

Explanation: Hard disk is present inside a computer. So, it is not a removable drive.

9. In ROM, each bit combination that comes out of the output lines is called _____

- a) Memory unit
- b) Storage class

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- c) Data word
- d) Address

Answer: c

Explanation: In ROM, each bit combination that comes out of the output lines is called data word. Usually, a word consists of 16-bits or 2-bytes.

10. VLSI chip utilizes _____

- a) NMOS
- b) CMOS
- c) BJT
- d) All of the Mentioned

Answer: d

Explanation: Very Large Scale Integration (VLSI) (ranging from 10,000 to 100,000 gates per IC) is a memory chip which is made up of NMOS, CMOS, BJT, and BiCMOS.

11. The MOS technology based semiconductor ROMs are classified into _____ categories.

- a) 2
- b) 3
- c) 4
- d) 5

Answer: b

Explanation: The MOS technology based semiconductor ROMs are classified into three categories: Mask ROM, PROM, & EPROM. PROM stands for Programmable Read Only Memory in which the ROM can be externally programmed by the user. EPROM stands for Erasable Programmable Read Only Memory, where the ROM can be cleared and re-programmed.

12. MOS ROM is constructed using _____

- a) FETs
- b) Transistors
- c) MOSFETs
- d) BJTs

Answer: c

Explanation: MOS ROM is made up of MOSFETs. MOSFETs are Metal Oxide Semiconductor Field Effect Transistors.

13. The full form of EEPROM is _____

- a) Erasable Electrically Programmable ROMs
- b) Electrically Erasable Programmable ROMs

- c) Electrically Erasable Programming ROMs
- d) Electrically Erasable Programmed ROMs

Answer: b

Explanation: The full form of EEPROM is Electrically Erasable Programmable ROMs. In EPROM (Erasable Programmable ROMs), the ROM is cleared by exposing it to UV radiation and also it's a tedious process. Whereas, in EEPROM, the ROM can be cleared electrically and thus is less time consuming and more efficient.

14. Which of the following best describes EPROMs?

- a) EPROMs can be programmed only once
- b) EPROMs can be erased by UV
- c) EPROMs can be erased by shorting all inputs to the ground
- d) EPROMs can be erased electrically

Answer: b

Explanation: EPROM (Erasable Programmable ROMs), the ROM is cleared by exposing it to UV radiation and also it's a tedious process. Whereas, in EEPROM, the ROM can be cleared electrically and thus is less time consuming and more efficient.

15. The Width of a processor's data path is measured in bits. Which of the following are common data paths?

- a) 8 bits
- b) 12 bits
- c) 16 bits
- d) 32 bits

Answer: a

Explanation: In generalised form, the data paths are of 8 bits. The data path, also known as data bus, is the channel through which the processor sends and receives data.

16. What type of memory is not directly addressable by the CPU and requires special software called EMS (expanded memory specification)?

- a) Extended
- b) Expanded
- c) Base
- d) Conventional

Answer: b

Explanation: Expanded memory is not directly addressable by the CPU. Expanded memory is the additional memory which is incorporated beyond the parent memory limit of the processor.

17. Which bus is used for input and output in case of microprocessor operation?

- a) Address bus
- b) System bus
- c) Control bus
- d) Data bus

View Answer

Answer: c

Explanation: The input and output are used to control the function of a microprocessor. Hence, the control bus is used to transfer the input and output signal from microprocessor to external peripherals and or from external peripherals to microprocessor.

18. What is the major difference between DRAM and SRAM?

- a) Dynamic RAMs are always active; static RAMs must reset between data read/write cycles
- b) SRAMs can hold data via a static charge, even with power off
- c) The only difference is the terminal from which the data is removed—from the FET Drain or Source
- d) DRAMs must be periodically refreshed

Answer: d

Explanation: DRAMs must be periodically refreshed so that it can store the new information. DRAMs are slower compared to SRAMs as the access time for SRAM is less than that of DRAM.

19. Which of the following is not a part of Hard disk?

- a) Platter
- b) Read/Write
- c) Valve
- d) Spindle

Answer: c

Explanation: A valve is a device that regulates, directs or controls the flow of a fluid (gases, liquids, fluidized solids, or slurries) by opening, closing, or partially obstructing various passageways. So, it is not a part of hard disk.

20. Which ROM can be erased by an electrical signal?

- a) ROM
- b) Mask ROM
- c) EPROM
- d) EEPROM

Answer: d

Explanation: In EPROM (Erasable Programmable ROMs), the ROM is cleared by exposing it

to UV radiation and also it's a tedious process. Whereas, in EEPROM, the ROM can be cleared electrically and thus is less time consuming and more efficient.

21. In the floppy drive, data is written to and read from the disk via a magnetic _____ head mechanism.

- a) Cluster
- b) Read/Write
- c) Cylinder
- d) Recordable

Answer: b

Explanation: A floppy disk is a removable disk used for storing data via magnetic facilities. In the floppy drive, data is written to and read from the disk via a magnetic read/write head mechanism.

22. What does the term "random access" mean in terms of memory?

- a) Any address can be accessed in systematic order
- b) Any address can be accessed in any order
- c) Addresses must be accessed in a specific order
- d) Any address can be accessed in reverse order

Answer: b

Explanation: "Random access" mean which can be accessed randomly and in other words any address can be accessed in any order.

23. Which type of ROM has to be custom built by the factory?

- a) EEPROM
- b) Mask ROM
- c) EPROM
- d) PROM

Answer: b

Explanation: All types of ROM are programmable and can be programmed as per requirement but the mask ROM is always programmed for specific application and it can't be reprogrammed. PROM stands for Programmable Read Only Memory in which the ROM can be externally programmed by the user. EPROM stands for Erasable Programmable Read Only Memory, where the ROM can be cleared and re-programmed.

24. The computer's main memory is _____

- a) Hard drive and RAM
- b) CD-ROM and hard drive
- c) RAM and ROM
- d) CMOS and hard drive

Answer: c

Explanation: The computer's main memory is RAM and ROM because all the storage related operation are performed by the data present in RAM/ROM. RAM stands for Random Access Memory where any address can accessed in any order. ROM stands for Read Only Memory wherefrom data can only be read.

25. A major disadvantage of the mask ROM is that _____

- a) It is time consuming to change the stored data when system requirements change
- b) It is very expensive to change the stored data when system requirements change
- c) It cannot be reprogrammed if stored data needs to be changed
- d) It has an extremely short life expectancy and requires frequent replacement

Answer: c

Explanation: A major disadvantage of the mask ROM is that it cannot be reprogrammed if stored data needs to be changed. PROM stands for Programmable Read Only Memory in which the ROM can be externally programmed by the user. EPROM stands for Erasable Programmable Read Only Memory, where the ROM and be cleared and re-programmed.

26. ROM may be programmed in _____ ways.

- a) 2
- b) 3
- c) 4
- d) 5

Answer: a

Explanation: ROM may be programmed in two different ways: (i) Mask Programming & (ii) PROM. Mask Programming is done by the manufacture. Whereas, PROM(Programmable ROM) is programmed by the user.

27. Which programming is done during the manufacturing process?

- a) Mask Programming
- b) PROM
- c) Both PROM and mask programming
- d) EPROM

Answer: a

Explanation: Mask ROM is permanently programmed during the manufacturing process. Whereas, PROM(Programmable ROM) is programmed by the user.

28. A photographic negative is called a _____

- a) Photo
- b) Negative

- c) Mask
- d) Virtual image

Answer: c

Explanation: A photographic negative is called a mask is used to control the electrical connections on the chip.

29. Mask programming is also known as _____

- a) EPROM
- b) PROM
- c) Custom programming
- d) Both PROM and EPROM

Answer: c

Explanation: Mask programming is also known as custom programming. Mask ROM is permanently programmed during the manufacturing process. Whereas, PROM(Programmable ROM) is programmed by the user.

30. The total storage capacity of $16 * 8$ ROM is _____

- a) 8 bits
- b) 16 bits
- c) 128 bits
- d) 64 bits

Answer: c

Explanation: ROM stands for Read Only Memory in which data is stored permanently and wherefrom data can only be read and rarely modified. The total storage capacity of $16 * 8$ ROM is 128 bits (i.e. $16 * 8 = 128$).

31. Which IC is a typical MSI/TTL based?

- a) IC 74187
- b) IC 74189
- c) IC 74188
- d) IC 74186

Answer: a

Explanation: MSI/TTL stands for Medium Scale Integration of Transistor-Transistor Logic. IC 74187 is a typical MSI/TTL based.

32. IC 74187 is of _____

- a) 512 bits
- b) 1024 bits
- c) 256 bits

d) 68 bits

Answer: b

Explanation: IC 74187 is of 1024 bits because it is organised as $256 * 4$. Thus, it has 256 rows and 4 columns.

33. How many rows and columns are present in IC 74187?

a) 128, 3

b) 128, 4

c) 256, 3

d) 256, 4

Answer: d

Explanation: IC 74187 is organised as $256 * 4$, hence it has 256 rows and 4 columns.

IC 74187 is of 1024 bits because it is organized as $256 * 4$.

34. Which of the following IC is of 256 bit?

a) IC 74187

b) IC 74189

c) IC 74188

d) IC 74186

Answer: c

Explanation: IC 74188 is of 256 bits. Since, it is organised as $32 * 8 = 256$. Thus, it has 32 rows and 8 columns.

35. Which IC is known as bipolar ROM?

a) IC 74187

b) IC 74189

c) IC 74188

d) IC 74186

Answer: c

Explanation: IC 74188 is known as bipolar ROM since it is made up of TTL logic.

36. How many address location a bipolar ROM has?

a) 16

b) 32

c) 64

d) 8

Answer: b

Explanation: Bipolar ROM means IC 74188 and it is organized as $32 * 8$. Thus, it has 32 rows and 8 columns. So, it has 32 address locations and each of which has 8 bits of storage.

37. Which of the following is known as MOS static ROM?

- a) TMS 45276
- b) TMS 45278
- c) TMS 45279
- d) TMS 45275

Answer: a

Explanation: TMS 45276 is known as MOS static ROM and it is made up of MOSFETs.

38. TMS 45276 is of _____

- a) 32 KB
- b) 56 KB
- c) 8 bits
- d) 4 bytes

Answer: a

Explanation: TMS 45276 is known as MOS static ROM and it is made up of MOSFETs. In ROM, the data remains even when the power is switched off. TMS 45276 is of 32 KB.

39. Which of the following has the capability to store the highest bits of data?

- a) TMS 45276
- b) IC 74188
- c) IC 74187
- d) IC 74185

Answer: a

Explanation: TMS 45276 has the capability to store the highest bits of data because of $32768 * 8 = 12,62,144$ organization.

40. What does CS mean in a chip?

- a) Storing Capacity
- b) Custom Select
- c) Chip Select
- d) Custom Storage

Answer: c

Explanation: CS means chip select and with the help of CS a chip is activated/deactivated. It is used for enabling or disabling the function of the chip.

41. ROMs are used to _____

- a) Store bootstrap program
- b) Character generation
- c) Code conversion
- d) All of the Mentioned

Answer: d

Explanation: ROM stands for Read Only Memory in which data is permanently stored, even when the power is turned off. It is used to store bootstrap program, character generation and code conversion.

2.) Programmable Read Only Memory

1. The time from the beginning of a read cycle to the end of tACS/tAA is called as _____

- a) Write enable time
- b) Data hold
- c) Read cycle time
- d) Access time

Answer: d

Explanation: The time from the beginning of a read cycle to the end of tACS/tAA is called as access time. It is the time in which data is fetched from the storage.

2. Why did PROM introduced?

- a) To increase the storage capacity
- b) To increase the address locations
- c) To provide flexibility
- d) To reduce the size

Answer: c

Explanation: In order to provide some flexibility in the possible applications of ROM, PROM is introduced. PROM stands for Programmable ROM, in which the ROM is programmed by the user.

3. Which of the following is programmed electrically by the user?

- a) ROM
- b) EPROM
- c) PROM
- d) EEPROM

Answer: c

Explanation: Programmable ROMs can be programmed electrically by the user but can't be reprogrammed. EEPROMs can be electrically erased and re-programmed by the user.

4. PROMs are available in _____

- a) Bipolar and MOSFET technologies
- b) MOSFET and FET technologies

- c) FET and bipolar technologies
- d) MOS and bipolar technologies

Answer: d

Explanation: PROMs (Programmable ROMs) can be programmed electrically by the user but can't be reprogrammed. PROMs are available in both bipolar and MOS (Metal Oxide Semiconductor) technologies.

5. The bit capacity of a memory that has 2048 addresses and can store 8 bits at each address is _____

- a) 4096
- b) 16384
- c) 32768
- d) 8129

Answer: b

Explanation: 1 address can store 8 bits. Therefore, total capacity of a memory having n addresses = $8 * n$.

Therefore, for 2048 addresses,

total capacity of a memory = $2048 * 8 = 16384$ bits.

6. How many $8k \times 1$ RAMs are required to achieve a memory with a word capacity of $8k$ and a word length of eight bits?

- a) Eight
- b) Two
- c) One
- d) Four

Answer: a

Explanation: RAM stands for Random Access Memory in which any memory address can be accessed in any order. It requires word of length 8 bits. So, one word needs of 1 bit and 8 bit requires 8 bits.

7. Which of the following best describes the fusible-link PROM?

- a) Manufacturer-programmable, reprogrammable
- b) Manufacturer-programmable, one-time programmable
- c) User-programmable, reprogrammable
- d) User-programmable, one-time programmable

Answer: d

Explanation: The fusible-link PROM is user programmable and one time programmable. It

means that a written program can not be reprogrammed. EPROMs can be erased and re-programmed.

8. How can ultraviolet erasable PROMs be recognized?

- a) There is a small window on the chip
- b) They will have a small violet dot next to the #1 pin
- c) Their part number always starts with a "U", such as in U12
- d) They are not readily identifiable, since they must always be kept under a small cover

Answer: a

Explanation: An ultraviolet erasable PROMs have small window on the chip with black marked. Such type of PROMS are called EPROMS which are cleared by exposing it to UV radiation. They are re-programmable.

9. Which part of a Flash memory architecture manages all chip functions?

- a) Program verify code
- b) Floating-gate MOSFET
- c) Command code
- d) Input/Output pins

Answer: b

Explanation: MOSFET technology is the best one in the manufacturing of chip because it has high flexibility and storage capacity. Thus, Floating-Gate MOSFET part of a Flash Memory architecture manages all chip functions.

10. How much locations an 8-bit address code can select in memory?

- a) 8 locations
- b) 256 locations
- c) 65,536 locations
- d) 131,072 locations

Answer: b

Explanation: An 8 bit address code requires 32 memory locations and it can hold maximum upto $32 * 8 = 256$ locations = 2^8 .

11. What is a fusing process?

- a) It is a process by which data is passed to the memory
- b) It is a process by which data is read through the memory
- c) It is a process by which programs are burnout to the diode/transistors
- d) It is a process by which data is fetched through the memory

Answer: c

Explanation: Fusing is a process by which programs are burnout to the diode/transistors and it can not be reprogrammed if any error occurs.

12. Fusing process is _____

- a) Reversible
- b) Irreversible
- c) Synchronous
- d) Asynchronous

Answer: b

Explanation: Since, any program cannot be reprogrammed in a PROM, so this process is irreversible as PROMs are programmed using the Fusing process. Fusing is a process by which programs are burnout to the diode/transistors and it can not be reprogrammed if any error occurs.

13. The cell type used inside a PROM is _____

- a) Link cells
- b) Metal cells
- c) Fuse cells
- d) Electric cells

Answer: c

Explanation: The cell type used inside a PROM is fuse cells by which a program is burnout. Fusing is a process by which programs are burnout to the diode/transistors and it can not be reprogrammed if any error occurs.

14. How many types of fuse technologies are used in PROMs?

- a) 2
- b) 3
- c) 4
- d) 5

Answer: b

Explanation: Fusing is a process by which programs are burnout to the diode/transistors and it can not be reprogrammed if any error occurs. Three types of fuse technologies are used in PROMs and these are: (i) Metal links, (ii) Silicon links, & (iii) p-n junctions.

15. Metal links are made up of _____

- a) Polycrystalline
- b) Magnesium sulphide
- c) Nichrome
- d) Silicon dioxide

Answer: c

Explanation: Metal links are made up of Nichrome materials.

16. Silicon links are made up of _____

- a) Polycrystalline silicon
- b) Polycrystalline magnesium
- c) Nichrome
- d) Silicon dioxide

Answer: a

Explanation: Metal links are made up of Nichrome materials. Silicon links are made up of polycrystalline silicon.

17. During programming p-n junction is _____

- a) Avalanche reverse biased
- b) Avalanche forward biased
- c) Zener reverse biased
- d) Zener reverse biased

Answer: a

Explanation: The sudden heavy flow of electrons in the reverse direction and heat cause aluminium ions to migrate. So, during programming p-n junction is avalanche reversed biased.

18. The full form of FAMOS is _____

- a) Floating Gate Avalanche Injection MOS
- b) Float Gate Avalanche Injection MOS
- c) Floating Gate Avalanche Induction MOS
- d) Float Gate Avalanche Induction MOS

Answer: a

Explanation: The full form of FAMOS is Floating Gate Avalanche Injection MOS. It is a floating gate transistor in which the trapped electrons is responsible for the dropping of the voltage.

19. PROM is programmed by _____

- a) EPROM programmer
- b) EEPROM programmer
- c) PROM programmer
- d) ROM programmer

Answer: c

Explanation: PROM is programmed by plugging it into a special device called PROM programmer. The ROM cannot be clear and hence PROM is a one-time programmable device.

20. The PROM starts out with _____

- a) 1s
- b) 0s
- c) Null
- d) Both 1s and 0s

Answer: b

Explanation: PROM is a one-time programmable device, which is programmed by the user. The PROM starts out with all 0s. These current pulses blow the fuse links, thus creating the desired pattern.

21. For the implementation of PROM, which IC is used?

- a) IC 74187
- b) IC 74186
- c) IC 74185
- d) IC 74184

Answer: b

Explanation: For implementation of PROM, IC 74186 is used. IC 74186 is of 512 bits ($62 * 8 = 512$). Thus, it has 62 rows and 8 columns.

22. IC 74186 is of _____

- a) 1024 bits
- b) 32 bits
- c) 512 bits
- d) 64 bits

Answer: c

Explanation: IC 74186 is of 512 bits ($62 * 8 = 512$). Thus, it has 62 rows and 8 columns.

23. How many memory locations are addressed using 18 address bits?

- a) 165,667
- b) 245,784
- c) 262,144
- d) 212,342

Answer: c

Explanation: For n address bits, the memory location will consist of 2^n bits. Using 18 address bits, $2^{18} = 262,144$ ($= 256 \text{ K}$) words are addressed.

24. How many address bits are needed to operate a $2\text{K} * 8$ -bit memory?

- a) 10
- b) 11

c) 12

d) 13

Answer: b

Explanation: For n address bits, the memory location will consist of 2^n bits. Thus, for 2K, only 11 address bits are required, because $2^{11} = 2K$.

25. What is the bit storage capacity of a ROM with a 1024×8 organization?

a) 1024

b) 4096

c) 2048

d) 8192

Answer: d

Explanation: For n address bits, the memory location will consist of 2^n bits. $1024 = 2^{10}$. So, $2^{10} * 2^3 = 1024 * 8 = 8192$ bit.

3.) Erasable Programmable Read Only Memory

1. EPROM uses an array of _____

a) p-channel enhancement type MOSFET

b) n-channel enhancement type MOSFET

c) p-channel depletion type MOSFET

d) n-channel depletion type MOSFET

Answer: b

Explanation: EPROMs are Erasable Programmable ROMs which can be erased using UV radiation and re-programmed. EPROM uses an array of n-channel enhancement type MOSFET with an insulated gate structure.

2. The EPROM was invented by _____

a) Wen Tsing Chow

b) Dov Frohman

c) Luis O Brian

d) J P Longwell

Answer: b

Explanation: The EPROM was invented by Dov Frohman of Intel in 1971. EPROMs are Erasable Programmable ROMs which can be erased using UV radiation and re-programmed.

3. Address decoding for dynamic memory chip control may also be used for _____

a) Chip selection and address location

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- b) Read and write control
- c) Controlling refresh circuits
- d) Memory mapping

Answer: a

Explanation: Address decoding for dynamic memory chip control may also be used for chip selection and address location. Chip Selection enables or disables the functioning of the chip.

4. Which of the following describes the action of storing a bit of data in a mask ROM?

- a) A 0 is stored by connecting the gate of a MOS cell to the address line
- b) A 0 is stored in a bipolar cell by shorting the base connection to the address line
- c) A 1 is stored by connecting the gate of a MOS cell to the address line
- d) A 1 is stored in a bipolar cell by opening the base connection to the address line

Answer: c

Explanation: The action of storing a bit of data in a mask ROM is that when a 1 is stored by connecting the gate of a MOS cell to the address line. Mask ROMs are programmed by the manufacturer and are custom made as per the user.

5. The check sum method of testing a ROM _____

- a) Allows data errors to be pinpointed to a specific memory location
- b) Provides a means for locating and correcting data errors in specific memory locations
- c) Indicates if the data in more than one memory location is incorrect
- d) Simply indicates that the contents of the ROM are incorrect

Answer: d

Explanation: If checking of a sum method goes wrong, it simply indicates that the contents of the ROM are incorrect.

6. The initial values in all the cells of an EPROM is _____

- a) 0
- b) 1
- c) Both 0 and 1
- d) Alternate 0s and 1s

Answer: b

Explanation: The initial values in all the cells of an EPROM is 1.

7. To store 0 in such a cell, the floating point must be _____

- a) Reprogrammed
- b) Restarted

- c) Charged
- d) Power off

Answer: c

Explanation: EPROMs are Erasable Programmable ROMs which can be erased using UV radiation and re-programmed. To store 0 in the cell of an EPROM, the floating point must be charged.

8. The major disadvantage of RAM is?

- a) Its access speed is too slow
- b) Its matrix size is too big
- c) It is volatile
- d) High power consumption

Answer: c

Explanation: RAM is volatile memory. Thus, RAM stores the data as long as it is powered on and once the power goes out, it loses its data.

9. Which one of the following is used for the fabrication of MOS EPROM?

- a) TMS 2513
- b) TMS 2515
- c) TMS 2516
- d) TMS 2518

Answer: c

Explanation: EPROMs are Erasable Programmable ROMs which can be erased using UV radiation and re-programmed. TMS 2516 is a MOS EPROM device.

10. How many addresses a MOS EPROM have?

- a) 1024
- b) 512
- c) 2516
- d) 256

Answer: c

Explanation: EPROMs are Erasable Programmable ROMs which can be erased using UV radiation and re-programmed. MOS EPROM (i.e. TMS 2516) has 2048 ($2^{11} = 2048$) addresses.

11. To read from the memory, the select input and the power down/program input must be

-
- a) HIGH
 - b) LOW

- c) Sometimes HIGH and sometimes LOW
- d) Alternate HIGH and LOW

Answer: b

Explanation: To read from the memory, the select input and the power down/program input must be LOW.

12. ROMs retain data when _____

- a) Power is on
- b) Power is off
- c) System is down
- d) All of the Mentioned

Answer: d

Explanation: ROM retains the data when power is off/on/down because it has to read the data from memory only and it is done in every condition. It is non-volatile memory.

13. Suppose that a certain semiconductor memory chip has a capacity of $8K \times 8$. How many bytes could be stored in this device?

- a) 8,000
- b) 65,536
- c) 8,192
- d) 64,000

Answer: c

Explanation: $8K = 8 * 1024 = 8192$.

14. When a RAM module passes the checker board test it is _____

- a) Able to read and write only 0s
- b) Faulty
- c) Probably good
- d) Able to read and write only 1s

Answer: c

Explanation: When a RAM module passes the checker board test it is probably good. It is a volatile memory. Thus, RAM stores the data as long as it is powered on and once the power goes out, it loses its data.

15. What is the difference between static RAM and dynamic RAM?

- a) Static RAM must be refreshed, dynamic RAM does not
- b) There is no difference
- c) Dynamic RAM must be refreshed, static RAM does not
- d) SRAM is slower than DRAM

Answer: c

Explanation: Dynamic RAM must be refreshed because it made up of capacitor, and capacitor required refresh. Static RAM made up of flip flop and it doesn't required a refresh.
